2023 AMC 10A Problem 21 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 21. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 21

Problem:

Let $P(x)$ be the unique polynomial of minimal degree with the following properties:

- $P(x)$ has leading coefficient 1,
- 1 is a root of $P(x) - 1$,
- 2 is a root of $P(x-2)$,
- 3 is a root of $P(3x)$, and
- 4 is a root of $4P(x)$.

The roots of $P(x)$ are integers, with one exception. The root that is not an integer can be written as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m+n$?

Answer Choices:

A. 41

B. 43

C. 45

D. 47

E. 49

Solution:

The given conditions imply that

$$P(1)=1, \quad P(2-2)=P(0)=0, \quad P(3\cdot 3)=P(9)=0, \quad 4P(4)=P(4)=0.$$

The polynomial $Q(x)=x(x-4)(x-9)$ has the required roots, but $Q(1)=24\neq 1$. Because the leading coefficient of $P(x)$ is 1 , it follows that $P(x)$ must have an additional factor. To minimize the degree of $P(x)$, that factor must be linear, say $x-a$. Thus $P(x)=(x-a)Q(x)$, and

$$1=P(1)=(1-a)\cdot 1\cdot(-3)\cdot(-8)=24(1-a).$$

Thus $a=\frac{23}{24}$, and the requested sum is $23+24=\text{(D)}\boxed{47}$.